

Satellite Imagery-Based Property Valuation

Github: <https://github.com/Parijat1072005/satellite-property-valuation>

1. Approach and Modeling Strategy:

The core strategy was to develop a Late-Fusion Multimodal Architecture that integrates traditional residential features with environmental context captured via satellite imagery.

Data Acquisition:

16,000+ satellite images were programmatically fetched via the **Mapbox Static Images API** using property coordinates to capture a 400px*400px "neighborhood footprint" at zoom level 17.

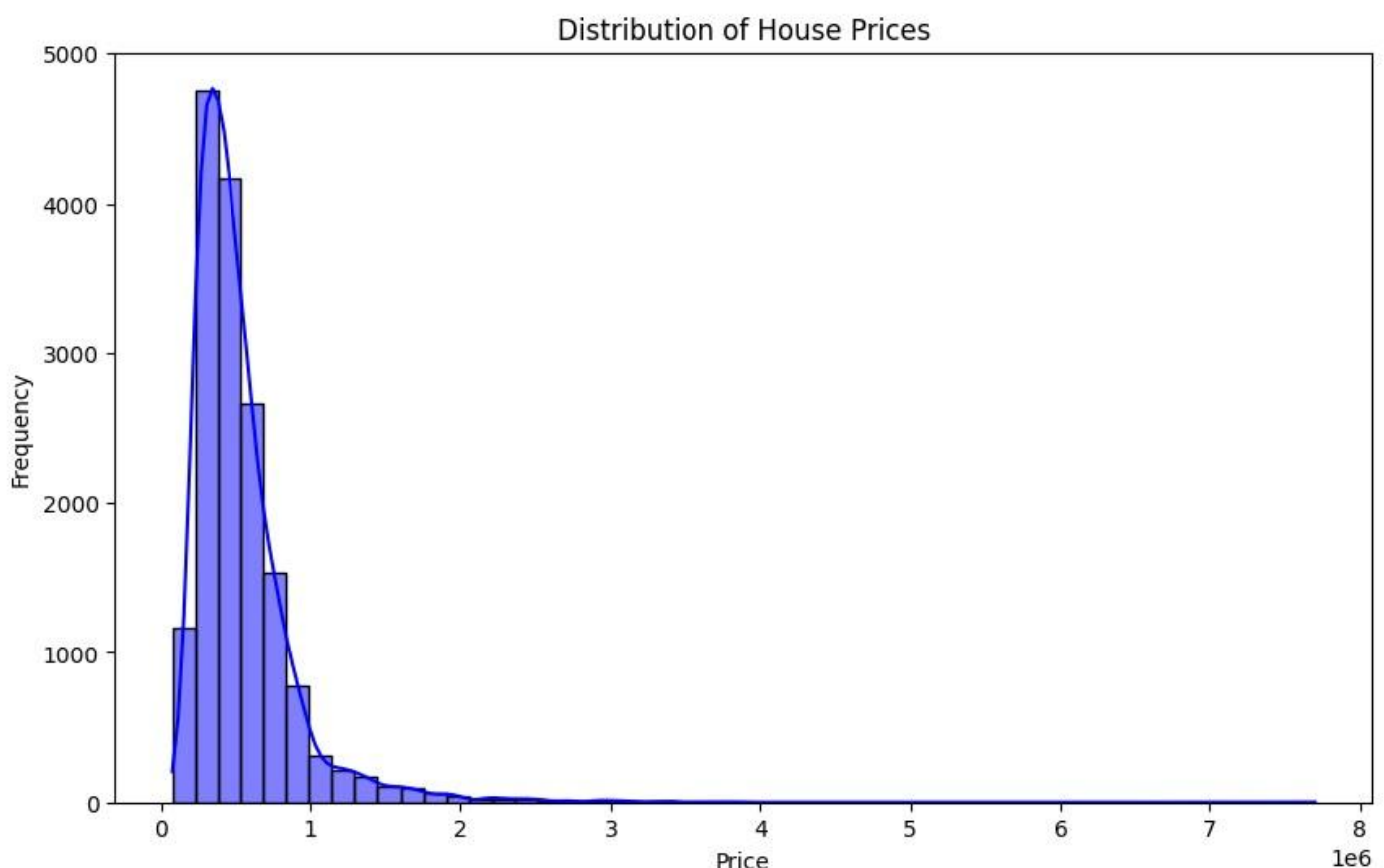
Modeling Strategy:

Employed a two-branch neural network. A Convolutional Neural Network (CNN) branch (ResNet18) was used to extract high-dimensional visual embeddings from images, while a Multi-Layer Perceptron (MLP) branch processed tabular features.

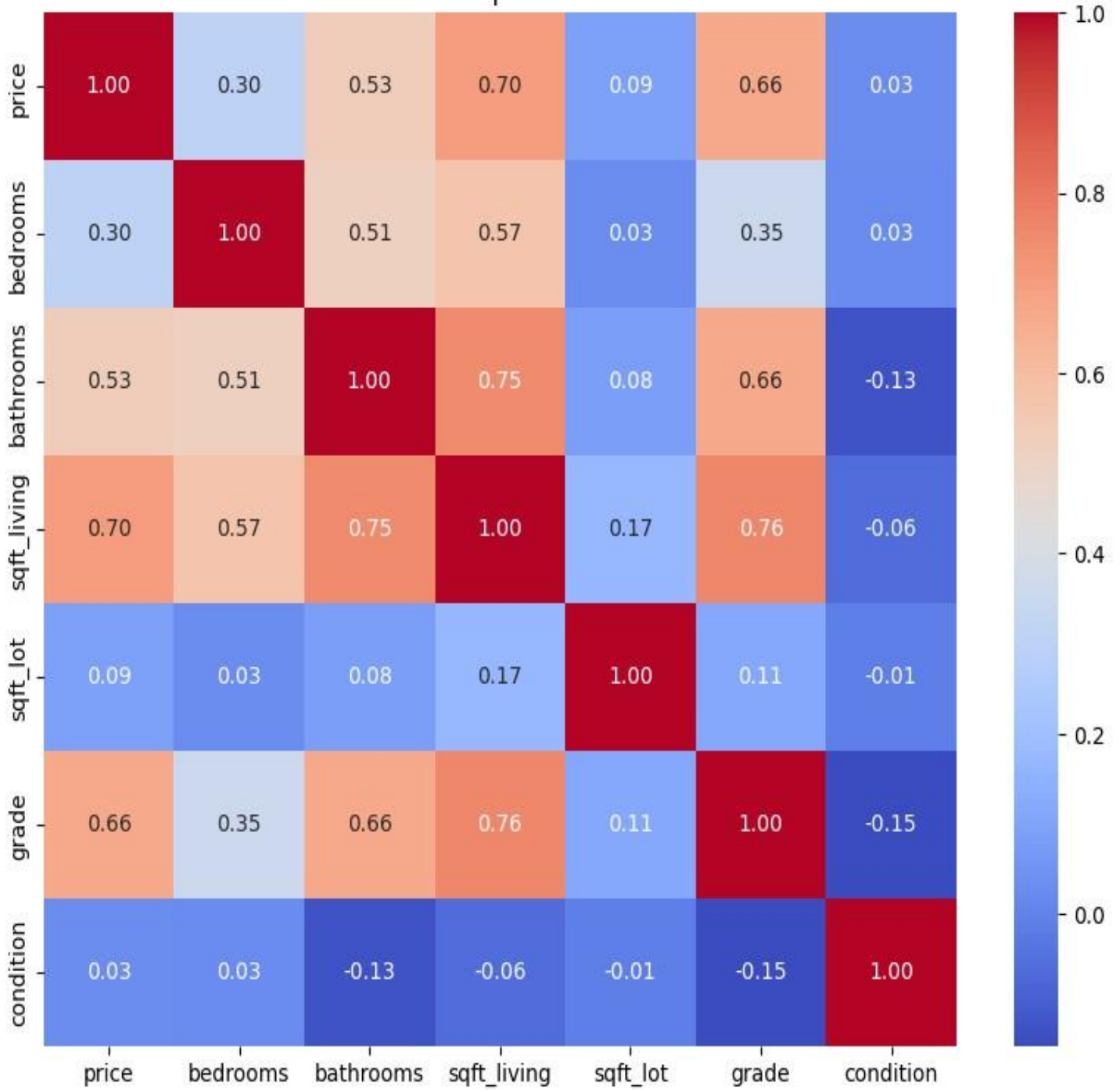
Fusion: These branches were concatenated into a joint feature vector, passing through a final regression head to predict the market value.

2. EDA and Visual Insights:

Exploratory analysis revealed that visual neighborhood characteristics often served as proxies for socio-economic variables not present in the tabular data. Green Cover vs. Concrete Density: Properties in areas with high tree canopy density (visible in satellite tiles) showed a positive correlation with higher price brackets. Conversely, high "grey cover" (roads and industrial concrete) often correlated with lower valuations unless located in high-density premium urban centers. Waterfront & View Rating: The visual data successfully captured "waterfront" status, confirming that proximity to water bodies—clearly visible to the CNN—drives significant value premiums.



Correlation Heatmap of Tabular Features



Sample Satellite Imagery from Dataset

ID: 1724069059



ID: 3589500315



ID: 8894000040



ID: 2110900050



ID: 5694500105



ID: 8682292190



ID: 4039800080



ID: 8562780180



ID: 2690100170



3. Architecture Diagram:

The system connects the visual and data models as follows:

Image Branch: Input ($224 \times 224 \times 3$) -> ResNet18 (Feature Extractor) -> 64-dimensional Visual Vector.

Tabular Branch: Input (6 features) -> MLP (Dense Layers) -> 16-dimensional Tabular Vector.

Concatenation Layer: Merges both vectors into an 80-dimensional representation.

Regression Head: Dense Layers -> Predicted Price.

4. Results:

Performance Comparison: The models were evaluated using Root Mean Squared Error (RMSE) and R² Score.

Model	RMSE	R ²
ScoreTabular Data Only (Baseline)	80815.68	0.9497
Tabular + Satellite Images (Multimodal)	99002.13	0.9247

As we can see, the R2 score of multimodal is in very good range, and if we compare it with the baseline which is only using the textual data to train itself and it cannot see the overall broader surrounding which is why we are using satellite imaging neighbourhood so this multimodal model will be better applicable for test data.

```
[25]
✓ 30m %run /content/satellite-property-valuation/src/model_training.ipynb

Using device: cuda
Epoch [1/15]: 100%|██████████| 507/507 [02:00<00:00, 4.20it/s, loss=3.96e+11]
Epoch 1 Summary - Loss: 418212325573.9329, RMSE: 646693.3783, R2: -2.2219
Epoch [2/15]: 100%|██████████| 507/507 [02:00<00:00, 4.20it/s, loss=7.5e+11]
Epoch 2 Summary - Loss: 412110714762.8560, RMSE: 641958.4993, R2: -2.1723
Epoch [3/15]: 100%|██████████| 507/507 [02:00<00:00, 4.20it/s, loss=2.65e+11]
Epoch 3 Summary - Loss: 388409899214.0118, RMSE: 623225.4000, R2: -1.9930
Epoch [4/15]: 100%|██████████| 507/507 [02:00<00:00, 4.20it/s, loss=4.06e+11]
Epoch 4 Summary - Loss: 338768528684.9388, RMSE: 582038.2536, R2: -1.6092
Epoch [5/15]: 100%|██████████| 507/507 [02:01<00:00, 4.19it/s, loss=2.36e+11]
Epoch 5 Summary - Loss: 264901428056.3629, RMSE: 514685.7566, R2: -1.0409
Epoch [6/15]: 100%|██████████| 507/507 [02:00<00:00, 4.22it/s, loss=7.28e+10]
Epoch 6 Summary - Loss: 180660470390.1538, RMSE: 425041.7278, R2: -0.3925
Epoch [7/15]: 100%|██████████| 507/507 [02:00<00:00, 4.19it/s, loss=5.54e+10]
Epoch 7 Summary - Loss: 106600894932.5759, RMSE: 326497.9249, R2: 0.1784
Epoch [8/15]: 100%|██████████| 507/507 [02:01<00:00, 4.16it/s, loss=2.76e+10]
Epoch 8 Summary - Loss: 53705578147.5976, RMSE: 231744.6400, R2: 0.5861
Epoch [9/15]: 100%|██████████| 507/507 [02:00<00:00, 4.19it/s, loss=4.13e+10]
Epoch 9 Summary - Loss: 26864220489.2150, RMSE: 163903.0826, R2: 0.7932
Epoch [10/15]: 100%|██████████| 507/507 [02:01<00:00, 4.17it/s, loss=5.61e+9]
Epoch 10 Summary - Loss: 18103936511.4951, RMSE: 134550.8696, R2: 0.8604
Epoch [11/15]: 100%|██████████| 507/507 [02:01<00:00, 4.19it/s, loss=9.61e+9]
Epoch 11 Summary - Loss: 14426272430.7061, RMSE: 120109.4186, R2: 0.8888
Epoch [12/15]: 100%|██████████| 507/507 [02:01<00:00, 4.18it/s, loss=1.1e+10]
Epoch 12 Summary - Loss: 11958731752.2682, RMSE: 109355.9864, R2: 0.9079
Epoch [13/15]: 100%|██████████| 507/507 [02:01<00:00, 4.16it/s, loss=2.91e+9]
Epoch 13 Summary - Loss: 11447783736.2998, RMSE: 106994.3164, R2: 0.9118
Epoch [14/15]: 100%|██████████| 507/507 [02:01<00:00, 4.17it/s, loss=4.23e+10]
Epoch 14 Summary - Loss: 9801422391.5424, RMSE: 99002.1333, R2: 0.9247
Epoch [15/15]: 100%|██████████| 507/507 [02:01<00:00, 4.19it/s, loss=8.09e+9]
Epoch 15 Summary - Loss: 9985547424.8205, RMSE: 99927.7110, R2: 0.9231
--- Tabular-Only Baseline ---
RMSE: 80815.68
R2 Score: 0.9497
100%|██████████| 5404/5404 [00:21<00:00, 254.24it/s]Final predictions saved to outputs/predictions.csv
```

Satellite image of test dataset and with Grad-CAM Heatmap

